



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)  
FULL SYLLABUS  
18-06-2022

## JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]

### ANSWER KEY

#### PART-1 : PHYSICS

| SECTION-I  | Q. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|------------|----|----|----|----|----|----|----|----|----|----|----|
|            | A. | B  | B  | C  | B  | A  | A  | B  | D  | C  | D  |
|            | Q. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| SECTION-II | A. | B  | C  | B  | B  | B  | C  | D  | D  | C  | C  |
|            | Q. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|            | A. | 9  | 2  | 1  | 60 | 2  | 30 | 1  | 2  | 10 | 6  |

#### PART-2 : CHEMISTRY

| SECTION-I  | Q. | 1  | 2  | 3   | 4    | 5  | 6  | 7   | 8  | 9   | 10 |
|------------|----|----|----|-----|------|----|----|-----|----|-----|----|
|            | A. | A  | B  | C   | D    | B  | C  | C   | D  | A   | B  |
|            | Q. | 11 | 12 | 13  | 14   | 15 | 16 | 17  | 18 | 19  | 20 |
| SECTION-II | A. | D  | C  | C   | A    | D  | D  | D   | D  | C   | B  |
|            | Q. | 1  | 2  | 3   | 4    | 5  | 6  | 7   | 8  | 9   | 10 |
|            | A. | 16 | 2  | 875 | 1575 | 10 | 6  | 125 | 60 | 300 | 28 |

#### PART-3 : MATHEMATICS

| SECTION-I  | Q. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8   | 9   | 10   |
|------------|----|----|----|----|----|----|----|----|-----|-----|------|
|            | A. | B  | D  | D  | C  | D  | A  | C  | C   | C   | B    |
|            | Q. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18  | 19  | 20   |
| SECTION-II | A. | C  | B  | A  | B  | B  | A  | B  | B   | D   | C    |
|            | Q. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8   | 9   | 10   |
|            | A. | 9  | 7  | 1  | 3  | 5  | 2  | 2  | 924 | 832 | 2020 |

### HINT – SHEET

#### PART-1 : PHYSICS

##### SECTION-I

##### 1. Ans (B)

Initially, energy of electron = +3eV  
finally, in 2<sup>nd</sup> excited state,

$$\text{energy of electron} = -\frac{(13.6\text{eV})}{3^2}$$

$$= -1.51\text{eV}$$

Loss in energy is emitted as photon,

$$\text{So, photon energy } \frac{hc}{\lambda} = 4.51\text{ eV}$$

Now, photoelectric effect equation

$$KE_{\max} = \frac{hc}{\lambda} - \phi = 4.51 - \left(\frac{hc}{\lambda_{\text{th}}}\right)$$

$$= 4.51\text{eV} - \frac{12400\text{ eV}\text{\AA}}{4000\text{\AA}}$$

$$= 1.41\text{ eV}$$

##### 2. Ans (B)

Standard graph between angle of deviation and incident angle.

##### 3. Ans (C)

At terminal speed

$$a = 0$$

$$F_{\text{net}} = 0$$

$$mg = F_v = 6\pi \eta Rv$$

$$v = \frac{mg}{6\pi \eta Rv}$$

$$v = \frac{\rho_w \frac{4\pi}{3} R^3 g}{6\pi \eta R}$$

$$= \frac{2\rho_w R^2 g}{9\eta}$$

$$= \frac{400}{81} \text{ m/s}$$

$$= 4.94 \text{ m/s}$$

4. **Ans (B)**

Since, each vibrational mode, corresponds to two degrees of freedom, hence,

$$f = 3 \text{ (trans.)} + 3 \text{ (rot.)} + 8 \text{ (vib.)} = 14$$

$$\& \gamma = 1 + \frac{2}{f}$$

$$\gamma = 1 + \frac{2}{14} = \frac{8}{7}$$

$$W = \frac{nR\Delta T}{\gamma - 1} = -582$$

As  $W < 0$ , work is done on the gas.

5. **Ans (A)**

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$0.2 = 2\pi \sqrt{\frac{0.5}{k}}$$

$$k = 50\pi^2$$

$$\approx 500$$

$$x = A \sin(\omega t + \phi)$$

$$= 5 \text{ cm} \sin\left(\frac{\omega T}{4} + 0\right)$$

$$= 5 \text{ cm} \sin\left(\frac{\pi}{2}\right)$$

$$= 5 \text{ cm}$$

$$PE = \frac{1}{2} kx^2$$

$$= \frac{1}{2} (500) \left(\frac{5}{100}\right)^2$$

$$= 0.6255$$

6. **Ans (A)**

$$q = CV$$

$$[C] = \left[\frac{q}{V}\right] = \frac{(A \times T)^2}{ML^2 T^{-2}}$$

$$= M^{-1} L^{-2} T^4 A^2$$

$$[E] = \left[\frac{F}{q}\right] = \frac{MLT^{-2}}{AT}$$

$$= MLT^{-3} A^{-1}$$

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$$

$$[\epsilon_0] = M^{-1} L^{-3} T^4 A^2$$

$$\text{Speed of light } c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\mu_0 = \frac{1}{\epsilon_0 c^2}$$

$$[\mu_0] = \frac{1}{[M^{-1} L^{-3} T^4 A^2][L T^{-1}]^2}$$

$$= [M^1 L^1 T^{-2} A^{-2}]$$

7. **Ans (B)**

$$E_{\text{mech.}} = 8J$$

(A) at  $x > x_4$ ,

$$U = \text{constant} = 6J$$

$$K = E_{\text{mech.}} - U = 2J = \text{constant}$$

(B) at  $x < x_1$ ,

$$U = \text{constant} = 8J$$

$$K = E_{\text{mech.}} - U = 8 - 8 = 0J$$

Particle is at rest.

(C) At  $x = x_2$ ,  $U = 0 \Rightarrow E_{\text{mech.}} = K = 8J$

KE is greatest, and particle is moving at fastest speed.

(D) At  $x = x_3$ ,  $U = 4J$

$$U + K = 8J$$

$$K = 4J$$

8. **Ans (D)**

$$C = 0.1 \mu F = 10^{-7} F$$

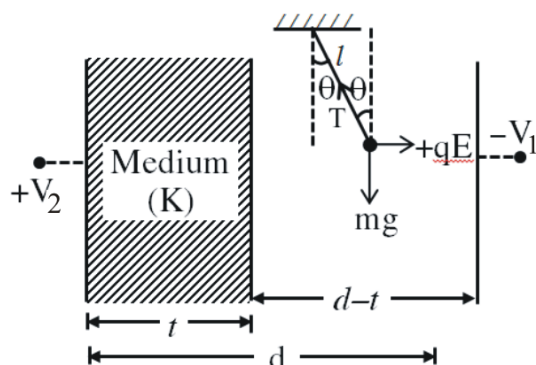
Resonant frequency = 60 Hz

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$2\pi f_0 = \frac{1}{\sqrt{LC}} \Rightarrow L = \frac{1}{4\pi^2 f_0^2 C}$$

by putting values  $L \simeq 70.3 \text{ Hz}$ .

9. Ans (C)

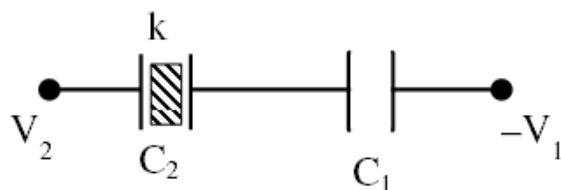


Let  $E$  be electric field in air

$$T \sin \theta = qE$$

$$T \cos \theta = mg$$

$$\tan \theta = \frac{qE}{mg}$$



$$Q = \left[ \frac{C_1 C_2}{C_1 + C_2} \right] [V_1 + V_2]$$

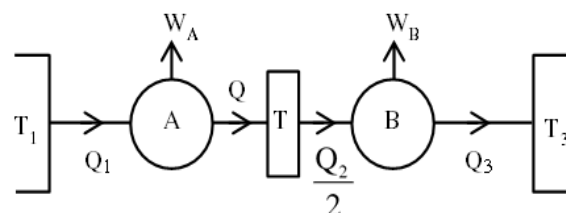
$$E = \frac{Q}{A \epsilon_0} = \left[ \frac{C_1 C_2}{C_1 + C_2} \right] \frac{[V_1 + V_2]}{A \epsilon_0}$$

$$C_1 = \frac{\epsilon_0 A}{d-t} \Rightarrow E = \frac{C_2 [V_1 + V_2]}{(C_1 + C_2)(d-t)}$$

$$\text{Now } \theta = \tan^{-1} \left[ \frac{q \cdot E}{mg} \right]$$

$$\theta = \tan^{-1} \left[ \frac{q}{mg} \times \frac{C_2 (V_1 + V_2)}{(C_1 + C_2)(d-t)} \right]$$

10. Ans (D)



$$W_A = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T}{T_1} \Rightarrow \frac{Q_2}{Q_1} = \frac{T}{T_1}$$

$$W_B = 1 - \frac{Q_3}{(Q_2/2)} = 1 - \frac{T_3}{T} \Rightarrow \frac{2Q_3}{Q_2} = \frac{T_3}{T}$$

$$\text{Now, } W_A = W_B$$

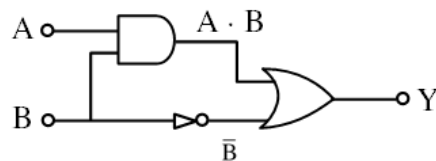
$$Q_1 - Q_2 = \frac{Q_2}{2} - Q_3$$

$$\Rightarrow \frac{2Q_1}{Q_2} + \frac{2Q_3}{Q_2} = 3$$

$$\Rightarrow \frac{2T_1}{T} + \frac{T_3}{T} = 3$$

$$\frac{2T_1}{3} + \frac{T_3}{3} = T$$

11. Ans (B)

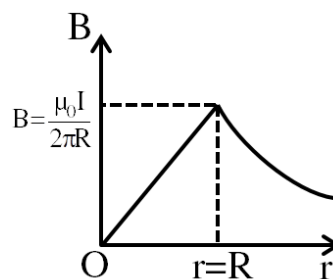


$$Y = A \cdot B + \bar{B}$$

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

12. Ans (C)

Graph for wire of radius  $R$  :



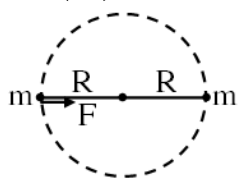
As  $b > a$

$$B_a > B_b$$

$$B_a = \frac{\mu_0 i}{2\pi a}$$

$$B_b = \frac{\mu_0 i}{2\pi b}$$

13. Ans (B)



$$F = \frac{Gm^2}{(2R)^2} = mR\omega^2$$

$$\omega = \frac{1}{2} \sqrt{\frac{G}{R^3}}$$

14. Ans (B)

(A) True, atom of each element emits characteristic spectrum.

(B) True, according to Bohr's postulates

$mvr = \frac{nh}{2\pi}$  and hence electron resides into orbits of specific radius called stationary orbits.

(C) False, density of nucleus is constant

(D) False, A free neutron is unstable decays into proton and electron and antineutrino.

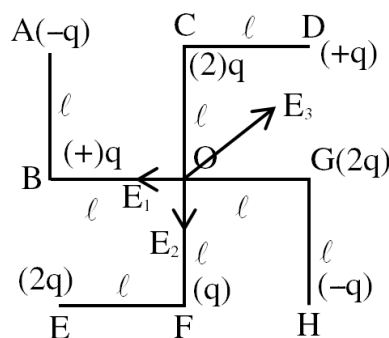
(E) True unstable nucleus show radioactivity.

15. Ans (B)

$$E_1 = \frac{kq}{\ell^2} = E_2$$

$$E_3 = \frac{kq}{(\sqrt{2}\ell)^2} = \frac{kq}{2\ell^2}$$

$$E = \frac{\sqrt{2}kq}{\ell^2} - \frac{kq}{2\ell^2} = \frac{kq}{2\ell^2} (2\sqrt{2} - 1)$$



16. Ans (C)

$$\frac{\Delta y}{y} = \frac{2\Delta m}{m} + \frac{4\Delta r}{r} + \frac{x\Delta g}{g} + \frac{3}{2} \frac{\Delta \ell}{\ell}$$

$$18 = 2(1) + 4(0.5) + xp + \frac{3}{2}(4)$$

$$8 = xp$$

By checking from options.

$$x = \frac{16}{3}, p = \pm \frac{3}{2}$$

17. Ans (D)

$$P = \text{const.}$$

$$P = Fv = \frac{mv^2 dv}{dx}$$

$$\int_0^x \frac{P}{m} dx = \int_0^v v^2 dv$$

$$\frac{Px}{m} = \frac{v^3}{3}$$

$$\left( \frac{3Px}{m} \right)^{1/3} = v = \frac{dx}{dt}$$

$$\left( \frac{3P}{m} \right)^{1/3} \int_0^t dt = \int_0^x x^{-1/3} dx$$

$$\Rightarrow x = \left( \frac{8P}{9m} \right)^{1/2} t^{3/2}$$

18. Ans (D)

Option D is correct

$$T^2 = \frac{4\pi^2}{GM} \cdot r^3$$

$$M = \frac{4\pi^2}{G} \cdot \frac{r^3}{T^2}$$

by putting values

$$M = 6 \times 10^{23}$$

19. Ans (C)

$$t = 0, u = 0$$

$$a = \frac{F_0}{M} - \frac{F_0}{MT^2} (t - T)^2 = \frac{dv}{dt}$$

$$\int_0^v dv = \int_{t=0}^{2T} \left( \frac{F_0}{M} - \frac{F_0}{MT^2} (t - T)^2 \right) dt$$

$$V = \left[ \frac{F_0}{M} t \right]_0^{2T} - \frac{F_0}{MT^2} \left[ \frac{t^3}{3} - t^2 T + T^2 t \right]_0^{2T}$$

$$V = \frac{4F_0 T}{3M}$$

20. Ans (C)

$$16 = R_0 [1 + \alpha (15 - T_0)]$$

$$20 = R_0 [1 + \alpha (100 - T_0)]$$

Assuming  $T_0 = 0^\circ\text{C}$ , as a general convention.

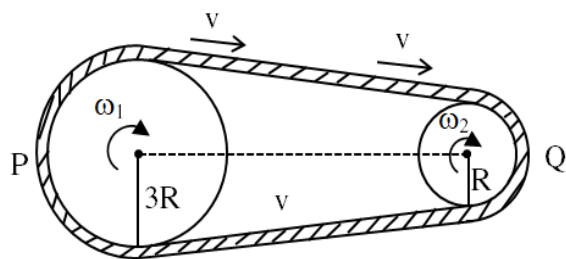
$$\Rightarrow \frac{16}{20} = \frac{1 + \alpha \times 15}{1 + \alpha \times 100}$$

$$\Rightarrow \alpha = 0.003^\circ\text{C}^{-1}$$

## PART-1 : PHYSICS

### SECTION-II

1. **Ans (9)**



$$\frac{1}{2} I_1 (\omega_1)^2 = \frac{1}{2} I_2 (\omega_2)^2$$

$$I_1 \left( \frac{v}{3R} \right)^2 = I_2 \left( \frac{v}{R} \right)^2$$

$$\frac{I_1}{I_2} = \frac{9}{1}$$

2. **Ans (2)**

Thickness  $t = n\lambda$

$$\text{So, } n \lambda_{\text{vac}} = (n+1) \lambda_{\text{air}}$$

$$n \lambda = (n+1) \frac{\lambda}{\mu_{\text{air}}}$$

$$n = \frac{1}{\mu_{\text{air}} - 1} = \frac{10^4}{3}$$

$$t = n\lambda$$

$$= \frac{10^4}{3} \times 6000 \text{ \AA}$$

$$= 2 \text{ mm}$$

3. **Ans (1)**

$$A_{\text{max}} = A_c + A_m = 12$$

$$A_{\text{min}} = A_c - A_m = 3$$

$$\Rightarrow A_c = \frac{15}{2} \text{ \& } A_m = \frac{9}{2}$$

$$\text{modulation index} = \frac{A_m}{A_c} = \frac{9/2}{15/2} = 0.6$$

$$\Rightarrow x = 1$$

4. **Ans (60)**

$$|\epsilon| = \frac{d\phi}{dt} = 20t + 20 \text{ mV}$$

$$|i| = \frac{|\epsilon|}{R} = 10t + 10 \text{ mA}$$

$$\text{at } t = 5$$

$$|i| = 60 \text{ mA}$$

5. **Ans (2)**

$$x(t) = A \sin(\omega t + \phi)$$

$$v(t) = A\omega \cos(\omega t + \phi)$$

$$2 = A \sin \phi \quad \dots\dots(1)$$

$$2\omega = A\omega \cos \phi \quad \dots\dots(2)$$

From (1) and (2)

$$\tan \phi = 1$$

$$\phi = 45^\circ$$

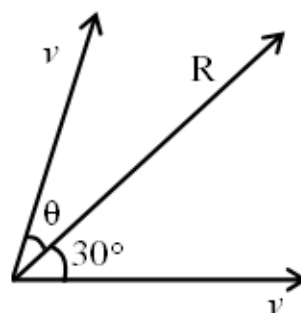
Putting value of  $\phi$  in equation (1)

$$2 = A \left\{ \frac{1}{\sqrt{2}} \right\}$$

$$A = 2\sqrt{2}$$

$$x = 2$$

6. **Ans (30)**

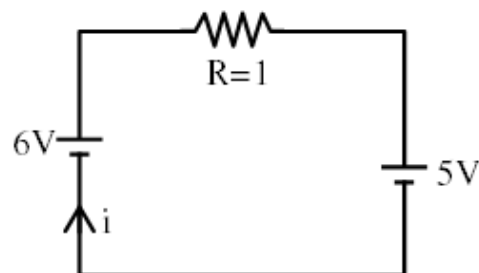


Both velocity vectors are of same magnitude therefore resultant would pass exactly midway through them

$$\theta = 30^\circ$$

7. **Ans (1)**

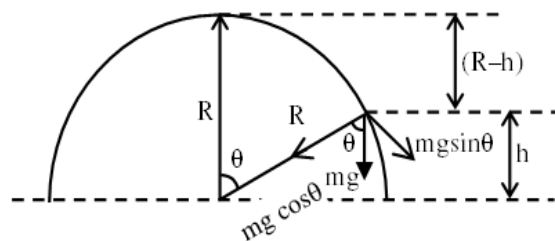
From graph voltage at  $t = 3.2$  sec is 6 volt.



$$i = \frac{6-5}{1}$$

$$i = 1 \text{ A}$$

8. Ans (2)



$$mg \cos \theta = \frac{mv^2}{R} \quad \dots(1)$$

$$\cos \theta = \frac{h}{R}$$

Energy conservation

$$mg \{R - h\} = \frac{1}{2} mv^2 \quad \dots(2)$$

$$\text{from (1) \& (2)} \Rightarrow mg \left\{ \frac{h}{R} \right\} = \frac{2mg \{R - h\}}{R}$$

$$h = \frac{2R}{3} = 2m$$

9. Ans (10)

$$E_{k\alpha} = E_k - E_L$$

$$\frac{hc}{\lambda_{k\alpha}} = E_k - E_L$$

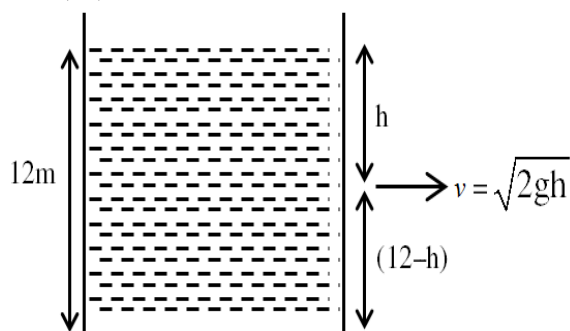
$$E_L = E_k - \frac{hc}{\lambda_{k\alpha}}$$

$$= 27.5 \text{ KeV} - \frac{12.42 \times 10^{-7} \text{ eV m}}{0.071 \times 10^{-9} \text{ m}}$$

$$E_L = (27.5 - 17.5) \text{ keV}$$

$$= 10 \text{ keV}$$

10. Ans (6)



$$R = \sqrt{2gh} \times \sqrt{\frac{(12-h) \times 2}{g}}$$

$$\sqrt{4h(12-h)} = R$$

For maximum R

$$\frac{dR}{dh} = 0$$

$$\Rightarrow h = 6m$$

## PART-2 : CHEMISTRY

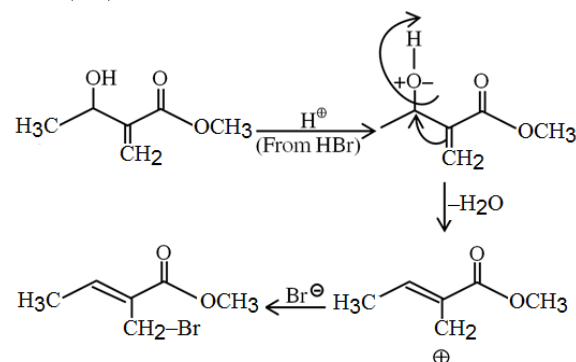
### SECTION-I

1. Ans (A)

By sodium fusion extract we can detect sulphur, nitrogen,

Phosphorous and halogens, because they are converted in to their ionic form with sodium metal.

2. Ans (B)



3. Ans (C)

Radioactive isotope of hydrogen is Tritium ( ${}^3_1\text{T}$ )

No. of neutrons ( $A-Z$ ) =  $3 - 1 = 2$

No. of electrons = 1

4. Ans (D)

Li makes alloy with Lead to make white metal bearings for motor engines

Liquid Na metal is used as coolant in fast breeder nuclear reactor

K is a very absorbent of  $\text{CO}_2$

Cs is used in making photoelectric cell

5. Ans (B)

Gases having higher critical temperature absorb to a greater extent.

6. Ans (C)

Mg Al P S  $\rightarrow$  IE. order  $\Rightarrow \text{Al} < \text{Mg} < \text{S} < \text{P}$

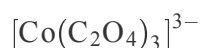
|                         |        |             |             |             |
|-------------------------|--------|-------------|-------------|-------------|
|                         | Mg     | Al          | P           | S           |
| Valence $[\text{Ne}] :$ | $3s^2$ | $3s^2 3p^1$ | $3s^2 3p^3$ | $3s^2 3p^4$ |
|                         |        | ↑           |             | ↑           |
|                         |        | Full        |             | Half        |
|                         |        | Filled      |             | Filled      |
|                         |        | Stable      |             | Stable      |

7. **Ans (C)**

**Statement I :** It is correct statement

**Statement II :**  $\text{CH}_3 - \overset{\oplus}{\text{CH}_2}$  involve  $\text{C}_{\text{sp}^3} - \text{H}_{1\text{s}}$  bond with empty 2p orbital hence given statement is false.

8. **Ans (D)**

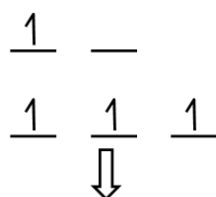


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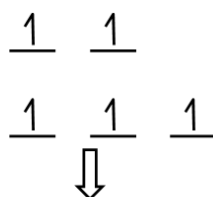
$\text{Mn}^{3+}, \text{CN}^-$      $\text{Fe}^{3+}, \text{CN}^-$      $\text{Co}^{3+}, \text{C}_2\text{O}_4^{2-}$   
 $d^4$  configuration, SFL    $d^5$  configuration, SFL    $d^6$  configuration, Chelating ligand

⇒ All will have larger splitting hence  $d^2\text{sp}^3$  hybridisation

$[\text{MnCl}_6]^{3-}$                       and                       $[\text{FeF}_6]^{3-}$   
 $d^4$  configuration,  $\text{Cl}^-$                        $d^5$  configuration,  $\text{F}^-$   
WFL                                              WFL



4 unpaired  
electrons



5 unpaired  
electrons

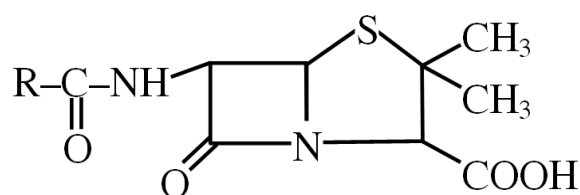
9. **Ans (A)**

$\text{Al}^{3+}$  and  $\text{Fe}^{3+}$  sulphides hydrolyse in water.  
 $\text{Ni}^{2+}$  and  $\text{Zn}^{2+}$  require basic medium with  $\text{H}_2\text{S}$  to form ppt  
 $\text{Ca}^{2+}$  and  $\text{Ba}^{2+}$  sulphides are soluble  
hence we will receive only  $\text{CuS}$  ppt.

10. **Ans (B)**

**Statement I :** Pencillin is bactericidal not bacteriostatic hence given statement is false.

**Statement II :** Structure of pencilline given is correct

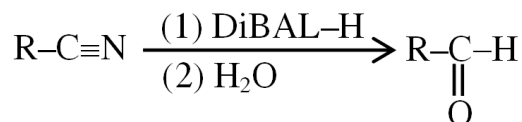


11. **Ans (D)**

**Lactose :** It is a disaccharide of  $\beta$ -D-Galactose and  $\beta$ -D-Glucose with  $\text{C}_1$  of galactose and  $\text{C}_4$  of glucose link.

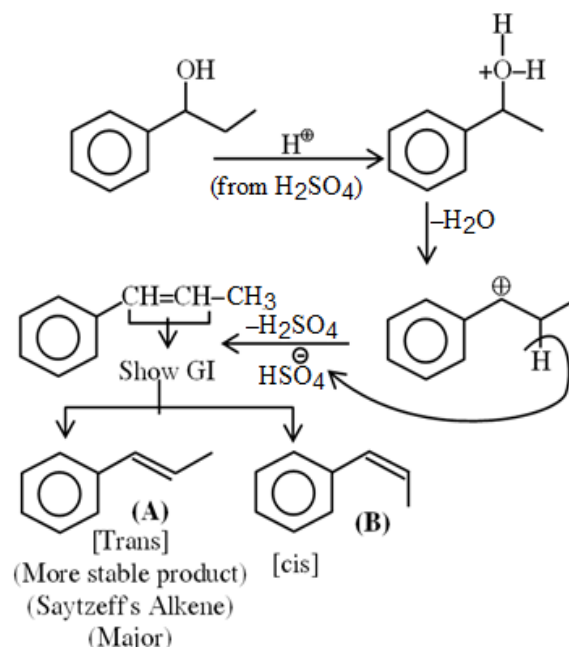
**Lactose :**  $\beta$ -D-Galactose +  $\beta$ -D-Glucose.

12. **Ans (C)**



Here Y is  $-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{H}$  Aldehyde

13. **Ans (C)**

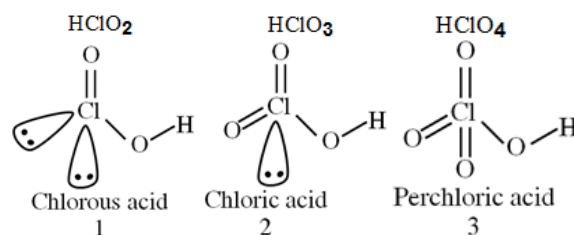


15. **Ans (D)**

As in Thomson model, protons are diffused (charge is not centred)  $\alpha$ - particles deviate by small angles and due to repulsion from protons, their speed decreases.

16. **Ans (D)**

Number of  $\text{Cl}=\text{O}$  bonds



## PART-2 : CHEMISTRY

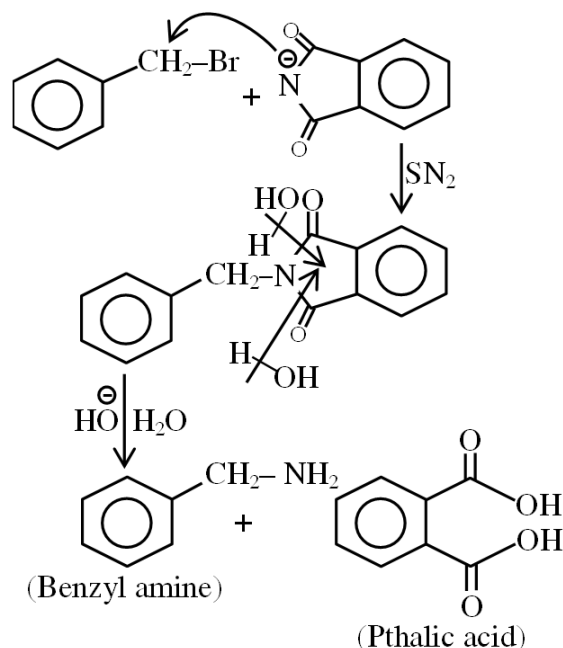
### SECTION-II

17. **Ans (D)**

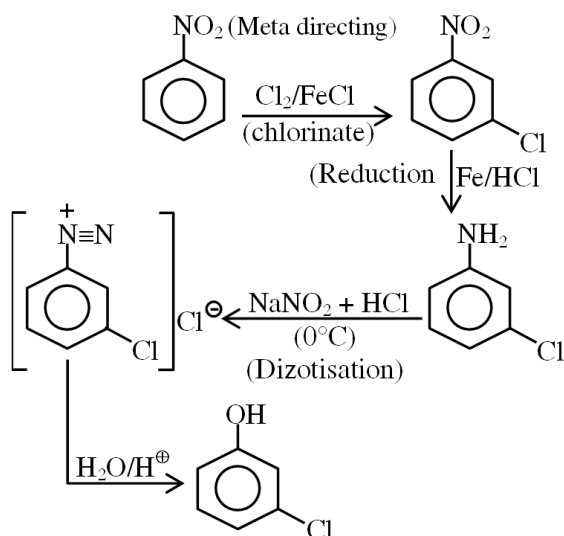
(A) Crystalline solids have definite arrangement of constituent particles and have long range order.

(C), (D) Different constituent particles of an amorphous solid have different bond strengths and soften over a range of temperatures.

18. **Ans (D)**

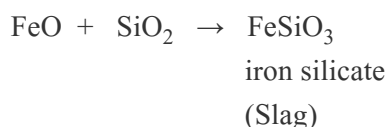


19. **Ans (C)**



20. **Ans (B)**

Silica is used to remove FeO impurity from the ore of copper



1. **Ans (16)**

$$k_p = P_{\text{O}_2}^{1/2} = 4$$

$$\therefore P_{\text{O}_2} = 16 \text{ bar} = 16 \text{ atm}$$

2. **Ans (2)**

$$n_{\text{H}^+} = \frac{400 \times 0.2}{1000} \times 2 = 0.16$$

$$n_{\text{OH}^-} = \frac{600 \times 0.1}{1000} = 0.06 \text{ (L.R.)}$$

Now, heat liberated from reaction

= heat gained by solutions

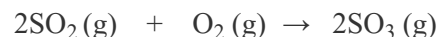
$$\text{or, } 0.06 \times 57.1 \times 10^3$$

$$= (1000 \times 1.0) \times 4.18 \times \Delta T$$

$$\therefore \Delta T = 0.8196 \text{ K}$$

$$= 81.96 \times 10^{-2} \text{ K} \approx 82 \times 10^{-2} \text{ K}$$

3. **Ans (875)**



Initial 250 m bar 750 m bar 0  
(L.R.)

Final -250 m bar -125 m bar 250 m bar

0 625 m bar 250 m bar

$$\therefore \text{Final total pressure} = 625 + 250 = 875 \text{ m bar}$$

4. **Ans (1575)**

$$n_{\text{eq}} \text{KMnO}_4 = n_{\text{eq}} \text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$$

$$\text{or, } \frac{10 \times 0.05}{1000} \times 5 = \frac{10 \times M}{1000} \times 2$$

$$\therefore \text{Conc. of oxalic acid solution} = 0.125 \text{ M}$$

$$= 0.125 \times 126 \text{ g/L} = 15.75 \text{ g/L}$$

$$= 1575 \times 10^{-2} \text{ g/L}$$

5. **Ans (10)**

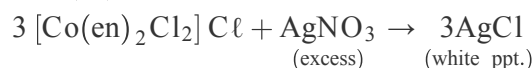
M. O. Configuration of  $\text{O}_2^{2-}$  ( $18\bar{e}$ )

$$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2$$

$$\pi^* 2p_x^2 = \pi^* 2p_y^2$$

Total B.M.O electrons = 10

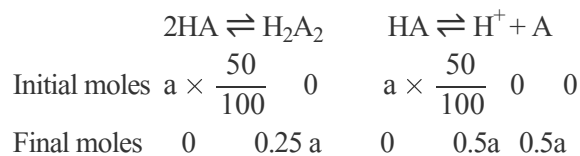
6. **Ans (6)**



Secondary valency of Co = 6

(C. N.)

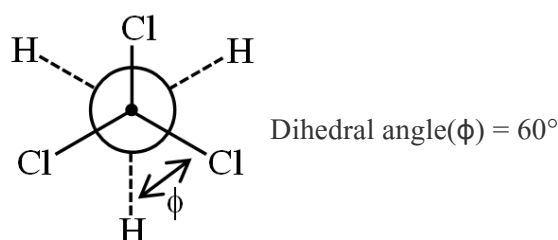
7. Ans (125)



$$\text{Now, } i = \frac{\text{final moles}}{\text{initial moles}} = \frac{0.25a + 0.5a + 0.5a}{0.5a + 0.5a} = 1.25 = 125 \times 10^{-2}$$

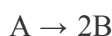
8. Ans (60)

1,1,1-Trichloro ethane [ $\text{CCl}_3\text{-CH}_3$ ]



(Newmonns stqquered form)

9. Ans (300)



|                       |                     |                     |
|-----------------------|---------------------|---------------------|
| $t = 0$               | 1 mole              | 0                   |
| $t = 100 \text{ min}$ | $1 - x$             | $2x$                |
|                       | $= 0.9 \text{ mol}$ | $= 0.2 \text{ mol}$ |

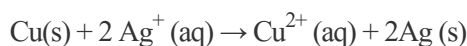
$$\text{Now, } t = \frac{t_{1/2}}{\ln 2} \times \frac{[A_0]}{[A_t]}$$

$$100 = \frac{t_{1/2}}{\ln 2} \times \ln \frac{1}{0.9} \Rightarrow t_{1/2} = 690 \text{ min.}$$

(taking  $\ln 3 = 1.11$ )

10. Ans (28)

Cell reaction is :



$$\text{Now, } E_{\text{cell}} = E_{\text{Cell}}^{\circ} - \frac{0.059}{2} \log \frac{[\text{Cu}^{2+}]}{[\text{Ag}^+]^2} \quad \dots (1)$$

$$\therefore E_1 = 0.3095 = E_{\text{Cell}}^{\circ} - \frac{0.059}{2} \cdot \log \frac{0.01}{(0.001)^2} \quad \dots (2)$$

$$\text{From (1) and (2), } E_2 = 0.28 \text{ V} = 28 \times 10^{-2} \text{ V}$$

## PART-3 : MATHEMATICS

### SECTION-I

1. Ans (B)

Image of  $A(a, b)$  along  $y = x$  is  $B(b, a)$ .

Translating it 2 units it becomes  $C(b + 2, a)$ .

Now, applying rotation theorem

$$-\frac{1}{2} + \frac{7}{\sqrt{2}}i = ((b + 2) + ai) \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$\frac{-1}{\sqrt{2}} + \frac{7}{\sqrt{2}}i = \left( \frac{b + 2}{\sqrt{2}} - \frac{a}{\sqrt{2}} \right) + i \left( \frac{b + 2}{\sqrt{2}} + \frac{a}{\sqrt{2}} \right)$$

$$\Rightarrow b - a + 2 = -1 \quad \dots (i)$$

$$\text{and } b + 2 + a = 7 \quad \dots (ii)$$

$$\Rightarrow a = 4; b = 1$$

$$\Rightarrow 2a + b = 9$$

2. Ans (D)

$${}^{10}\text{C}_8 \left( 25^{(x-1)} + 7 \right) \times \left( 5^{(x-1)} + 1 \right)^{-1} = 180$$

$$\Rightarrow \frac{25^{x-1} + 7}{5^{(x-1)} + 1} = 4$$

$$\Rightarrow \frac{t^2 + 7}{t + 1} = 4;$$

$$\Rightarrow t = 1, 3 = 5^{x-1}$$

$$\Rightarrow x - 1 = 0 \text{ (one of the possible value).}$$

$$\Rightarrow x = 1$$

3. Ans (D)

First line is  $(\phi + \alpha, 2\phi + 1, 3\phi + 1)$

and second line is  $(q\phi + 4, 3q + 6, 3q + 7)$ .

$$\text{For intersection } \phi + \alpha = q\phi + 4 \quad \dots (i)$$

$$2\phi + 1 = 3q + 6 \quad \dots (ii)$$

$$3\phi + 1 = 3q + 7 \quad \dots (iii)$$

$$\text{for (ii) \& (iii) } \phi = 1, q = -1$$

$$\text{So, from (i) } \alpha + \beta = 3$$

Now, point of intersection is  $(\alpha + 1, 3, 4)$

It lies on the plane.

$$\text{Hence, } \alpha = 5 \text{ \& } \beta = -2$$

4. Ans (C)

$$f(x) = \cos \lambda x$$

$$\therefore f\left(\frac{1}{2}\right) = -1$$

$$\text{So, } -1 = \cos \frac{\lambda}{2}$$

$$\Rightarrow \lambda = 2\pi$$

$$\text{Thus } f(x) = \cos 2\pi x$$

Now k is natural number

$$\text{Thus } f(k) = 1$$

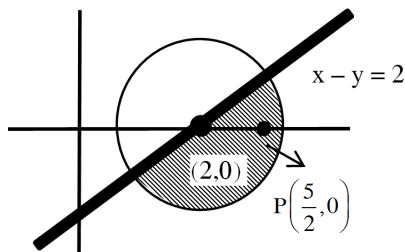
$$\sum_{k=1}^{20} \frac{1}{\sin k \sin(k+1)} = \frac{1}{\sin 1} \sum_{k=1}^{20} \left[ \frac{\sin((k+1) - k)}{\sin k \cdot \sin(k+1)} \right]$$

$$= \frac{1}{\sin 1} \sum_{k=1}^{20} (\cot k - \cot(k+1))$$

$$= \frac{\cot 1 - \cot 21}{\sin 1} = \operatorname{cosec}^2 1 \operatorname{cosec}(21) \cdot \sin 20$$

5. Ans (D)

$$|t - 2| \leq 1 \quad \text{Put } t = x + iy$$



$$(x - 2)^2 + y^2 \leq 1$$

$$\text{Also, } t(1 + i) + \bar{t}(1 - i) \geq 4$$

$$\text{Gives } x - y \geq 2$$

Let point on circle be  $A(2 + \cos \theta, \sin \theta)$

$$\theta \in \left[-\frac{3\pi}{4}, \frac{\pi}{4}\right]$$

$$(AP)^2 = \left(2 + \cos \theta - \frac{5}{2}\right)^2 + \sin^2 \theta$$

$$= \cos^2 \theta - \cos \theta + \frac{1}{4} + \sin^2 \theta$$

$$= \frac{5}{4} - \cos \theta$$

$$\text{For } (AP)^2 \text{ maximum } \theta = -\frac{3\pi}{4}$$

$$(AP)^2 = \frac{5}{4} + \frac{1}{\sqrt{2}} = \frac{5\sqrt{2} + 4}{4\sqrt{2}}$$

6. Ans (A)

$$P(E) < \frac{1}{2}$$

$$\Rightarrow \sum_{r=n}^8 {}^8C_r \left(\frac{1}{2}\right)^{8-r} \left(\frac{1}{2}\right)^r < \frac{1}{2}$$

$$\Rightarrow \sum_{r=n}^8 {}^8C_r \left(\frac{1}{2}\right)^8 < \frac{1}{2}$$

$$\Rightarrow {}^8C_n + {}^8C_{n+1} + \dots + {}^8C_8 < 128$$

$$\Rightarrow 256 - ({}^8C_0 + {}^8C_1 + \dots + {}^8C_{n-1}) < 128$$

$$\Rightarrow {}^8C_0 + {}^8C_1 + \dots + {}^8C_{n-1} > 128$$

$$\Rightarrow n - 1 \geq 4$$

$$\Rightarrow n \geq 5$$

7. Ans (C)

$$x = \frac{1}{2} \left( \tan \frac{\pi}{9} + \tan \frac{7\pi}{18} \right)$$

$$\text{and } 2y = \tan \frac{\pi}{9} + \tan \frac{5\pi}{18}$$

$$\text{so, } x - 2y = \frac{1}{2} \left( \tan \frac{\pi}{9} + \tan \frac{7\pi}{18} \right)$$

$$- \left( \tan \frac{\pi}{9} + \tan \frac{5\pi}{18} \right)$$

$$\Rightarrow |x - 2y| = \left| \frac{\cot \frac{\pi}{9} - \tan \frac{\pi}{9}}{2} - \tan \frac{5\pi}{18} \right|$$

$$= \left| \cot \frac{2\pi}{9} - \cot \frac{2\pi}{9} \right| = 0$$

$$\left( \text{as } \tan \frac{5\pi}{18} = \cot \frac{2\pi}{9}; \tan \frac{7\pi}{18} = \cot \frac{\pi}{9} \right)$$

8. Ans (C)

$$\text{Given } 32 + 8\alpha + 9\beta = (8 + \alpha + \beta) \times 6$$

$$\Rightarrow 2\alpha + 3\beta = 16 \quad \dots(i)$$

$$\text{Also, } 4 \times 16 + 4 \times \alpha + 9\beta = (8 + \alpha + \beta) \times 6.8$$

$$\Rightarrow 640 + 40\alpha + 90\beta = 544 + 68\alpha + 68\beta$$

$$\Rightarrow 28\alpha - 22\beta = 96$$

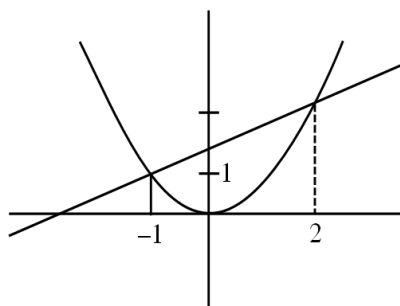
$$\Rightarrow 14\alpha - 11\beta = 48 \quad \dots(ii)$$

from (i) & (ii)

$$\alpha = 5 \text{ \& } \beta = 2$$

$$\text{so, new mean} = \frac{32 + 35 + 18}{15} = \frac{85}{15} = \frac{17}{3}$$

9. Ans (C)



$$y - x = 2, x^2 = y$$

$$\text{Now, } x^2 = 2 + x$$

$$\Rightarrow x^2 - x - 2 = 0$$

$$\Rightarrow (x + 1)(x - 2) = 0$$

$$\text{Area} = \int_{-1}^2 (2 + x - x^2)$$

$$= \left[ 2x + \frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^2$$

$$= \left( 4 + 2 - \frac{8}{3} \right) - \left( -2 + \frac{1}{2} + \frac{1}{3} \right)$$

$$= 6 - 3 + 2 - \frac{1}{2} = \frac{9}{2}$$

10. Ans (B)

$$(x - x^3)dy = (y + yx^2 - 3x^4)dx$$

$$\Rightarrow xdy - ydx = (yx^2 - 3x^4)dx + x^3dy$$

$$\Rightarrow \frac{xdy - ydx}{x^2} = (ydx + xdy) - 3x^2dx$$

$$\Rightarrow d\left(\frac{y}{x}\right) = d(xy) - d(x^3)$$

Integrate

$$\Rightarrow \frac{y}{x} = xy - x^3 + c$$

$$\text{given } f(3) = 3$$

$$\Rightarrow \frac{3}{3} = 3 \times 3 - 3^3 + c$$

$$\Rightarrow c = 19$$

$$\therefore \frac{y}{x} = xy - x^3 + 19$$

$$\text{at } x = 4, \frac{y}{4} = 4y - 64 + 19$$

$$15y = 4 \times 45$$

$$\Rightarrow y = 12$$

11. Ans (C)

$$\lim_{x \rightarrow 0} \left( \frac{x}{\sqrt[8]{1 - \sin x} - \sqrt[8]{1 + \sin x}} \right)$$

$$= \lim_{x \rightarrow 0} \left( \frac{x}{\sqrt[8]{1 - \sin x} - \sqrt[8]{1 + \sin x}} \right)$$

$$= \lim_{x \rightarrow 0} \left( \frac{x}{\sqrt[8]{1 - \sin x} - \sqrt[8]{1 + \sin x}} \right)$$

$$\left( \frac{(\sqrt[8]{1 - \sin x} + \sqrt[8]{1 + \sin x})}{\sqrt[8]{1 - \sin x} + \sqrt[8]{1 + \sin x}} \right)$$

$$\left( \frac{(\sqrt[4]{1 - \sin x} + \sqrt[4]{1 + \sin x})}{\sqrt[4]{1 - \sin x} + \sqrt[4]{1 + \sin x}} \right)$$

$$\left( \frac{(\sqrt[2]{1 - \sin x} + \sqrt[2]{1 + \sin x})}{\sqrt[2]{1 - \sin x} + \sqrt[2]{1 + \sin x}} \right)$$

$$= \lim_{x \rightarrow 0} \left( \frac{x}{1 - \sin x - (1 + \sin x)} \right)$$

$$(\sqrt[8]{1 - \sin x} + \sqrt[8]{1 + \sin x})$$

$$(\sqrt[4]{1 - \sin x} + \sqrt[4]{1 + \sin x})$$

$$(\sqrt[2]{1 - \sin x} + \sqrt[2]{1 + \sin x})$$

$$= \lim_{x \rightarrow 0} \frac{x}{(-2 \sin x)} (\sqrt[8]{1 - \sin x} + \sqrt[8]{1 + \sin x})$$

$$(\sqrt[4]{1 - \sin x} + \sqrt[4]{1 + \sin x})$$

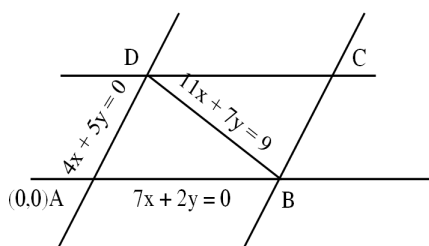
$$(\sqrt[2]{1 - \sin x} + \sqrt[2]{1 + \sin x})$$

$$= \lim_{x \rightarrow 0} \left( -\frac{1}{2} \right) (2) (2) (2)$$

$$\left\{ \because \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right\} = -4$$

12. Ans (B)

Both the lines pass through origin.



point D is equal of intersection of  $4x + 5y = 0$  &  $11x + 7y = 9$

So, coordinates of point D =  $\left(\frac{5}{3}, -\frac{4}{3}\right)$

Also, point B is point of intersection of  $7x + 2y = 0$  &  $11x + 7y = 9$

So, coordinates of point B =  $\left(-\frac{2}{3}, \frac{7}{3}\right)$

diagonals of parallelogram intersect at middle let middle point of B,D

$$\Rightarrow \left(\frac{\frac{5}{3} - \frac{2}{3}}{2}, \frac{-\frac{4}{3} + \frac{7}{3}}{2}\right) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

equation of diagonal AC

$$\Rightarrow (y - 0) = \frac{\frac{1}{2} - 0}{\frac{1}{2} - 0}(\pi - 0)$$

$$y = x$$

diagonal AC passes through (2, 2).

13. Ans (A)

$$\alpha = \max \{8^{2 \sin 3x} \cdot 4^{4 \cos 3x}\}$$

$$= \max \{2^{6 \sin 3x} \cdot 2^{8 \cos 3x}\}$$

$$= \max \{2^{6 \sin 3x + 8 \cos 3x}\}$$

$$\text{and } \beta = \min \{8^{2 \sin 3x} \cdot 4^{4 \cos 3x}\}$$

$$= \min \{2^{6 \sin 3x + 8 \cos 3x}\}$$

Now range of  $6 \sin 3x + 8 \cos 3x$

$$= \left[-\sqrt{6^2 + 8^2}, +\sqrt{6^2 + 8^2}\right] = [-10, 10]$$

$$\alpha = 2^{10} \text{ \& } \beta = 2^{-10}$$

$$\text{So, } \alpha^{1/5} = 2^2 = 4$$

$$\Rightarrow \beta^{1/5} = 2^{-2} = 1/4$$

quadratic  $8x^2 + bx + c = 0$ ,

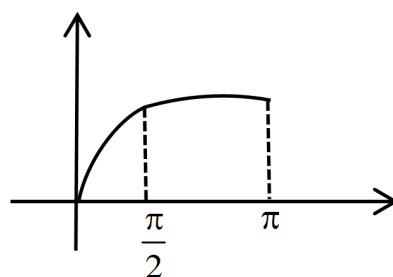
$$c - b = 8 \times [(\text{product of roots}) + (\text{sum of roots})]$$

$$= 8 \times \left[4 \times \frac{1}{4} + 4 + \frac{1}{4}\right]$$

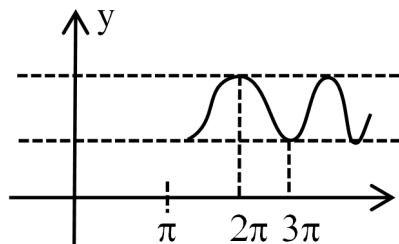
$$= 8 \times \left[\frac{21}{4}\right] = 42$$

14. Ans (B)

Graph of  $\max \{ \sin t : 0 \leq t \leq x \}$  in  $x \in [0, \pi]$

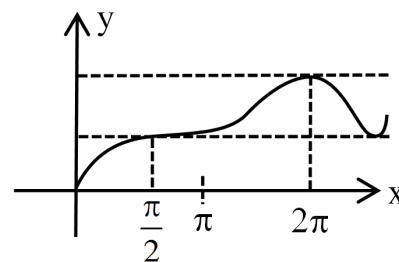


& graph of cos for  $x \in [\pi, \infty)$



So graph of

$$f(x) = \begin{cases} \max \{ \sin t : 0 \leq t \leq x, & 0 \leq x \leq \pi \\ 2 + \cos x & x > \pi \end{cases}$$



$f(x)$  is differentiable everywhere in  $(0, \infty)$

15. Ans (B)

$$x^3 - 3x^2y - xy^2 + 3y^3 = 0$$

$$\Rightarrow x(x^2 - y^2) - 3y(x^2 - y^2) = 0$$

$$\Rightarrow (x - 3y)(x - y)(x + y) = 0$$

Now,  $x = y \forall (x, y) \in \mathbb{N} \times \mathbb{N}$  so reflexive

But not symmetric & transitive

See, (3,1) satisfies but (1,3) does not. Also

(3,1) & (1,-1) satisfies but (3,-1) does not

16. Ans (A)

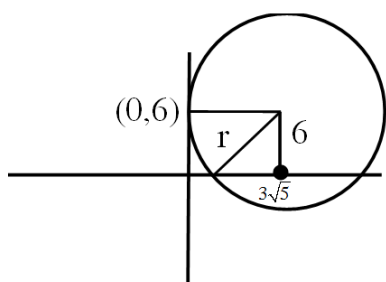
$P$  : for all  $M > 0$ , there exists  $x \in S$  such that  $x \geq M$ .

$\sim P$  : there exists  $M > 0$ , for all  $x \in S$

Such that  $x < M$

Negation of 'there exists' is 'for all'.

17. Ans (B)



$$r = \sqrt{6^2 + (3\sqrt{5})^2}$$

$$= \sqrt{36 + 45} = 9$$

18. Ans (B)

$$\vec{a} = (\vec{b} \cdot \vec{c}) \vec{b} - (\vec{b} \cdot \vec{b}) \vec{c}$$

$$= 1.2 \cos \theta \vec{b} - \vec{c}$$

$$\Rightarrow \vec{a} = 2 \cos \theta \vec{b} - \vec{c}$$

$$|\vec{a}|^2 = (2 \cos \theta)^2 + 2^2 - 2 \cdot 2 \cos \theta \cdot \vec{b} \cdot \vec{c}$$

$$\Rightarrow 2 = 4 \cos^2 \theta + 4 - 4 \cos \theta \cdot 2 \cos \theta$$

$$\Rightarrow -2 = -4 \cos^2 \theta$$

$$\Rightarrow \cos^2 \theta = \frac{1}{2}$$

$$\Rightarrow \sec^2 \theta = 2$$

$$\Rightarrow \tan^2 \theta = 1$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

$$1 + \tan \theta = 2.$$

19. Ans (D)

$$C = A^2 - B^2; |C| \neq 0$$

$$A^5 = B^5 \text{ and } A^3 B^2 = A^2 B^3$$

$$\text{Now, } A^5 - A^3 B^2 = B^5 - A^2 B^3$$

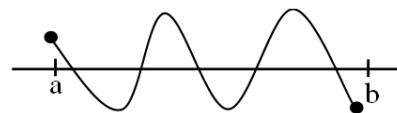
$$\Rightarrow A^3 (A^2 - B^2) + B^3 (A^2 - B^2) = 0$$

$$\Rightarrow (A^3 + B^3) (A^2 - B^2) = 0$$

$$\text{Post multiplying inverse of } A^2 - B^2:$$

$$A^3 + B^3 = 0$$

20. Ans (C)



$$f(x) = \int_a^x g(t) dt$$

$$f(x) \rightarrow 5$$

$$f'(x) \rightarrow 4$$

$$g(x) \rightarrow 4$$

$$g'(x) \rightarrow 3$$

## PART-3 : MATHEMATICS

### SECTION-II

1. Ans (9)

$$\vec{a} = (1, -\alpha, \beta)$$

$$\vec{b} = (3, \beta, -\alpha)$$

$$\vec{c} = (-\alpha, -2, 1); \alpha, \beta \in \mathbb{R}$$

$$\vec{a} \cdot \vec{b} = -1 \Rightarrow 3 - \alpha\beta - \alpha\beta = -1$$

$$\Rightarrow \alpha\beta = 2$$

$$\begin{vmatrix} 1 & 2 \\ 2 & 1 \\ -1 & -2 \\ -2 & -1 \end{vmatrix}$$

$$\vec{b} \cdot \vec{c} = 10$$

$$\Rightarrow -3\alpha - 2\beta - \alpha = 10$$

$$\Rightarrow 2\alpha + \beta + 5 = 0$$

$$\therefore \alpha = -2; \beta = -1$$

$$[\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 2 & -2 & 1 \end{vmatrix}$$

$$= 1(-1 + 4) - 2(3 - 4) - 1(-6 + 2)$$

$$= 3 + 2 + 4 = 9$$

2. Ans (7)

$$\overrightarrow{QR} : -\frac{x-3}{1} = \frac{y+4}{-1} = \frac{z+5}{-6} = r$$

$$\Rightarrow (x, y, z) \equiv (r+3, -r-4, -6r-5)$$

$$\text{Now, satisfying it in the given plane.}$$

$$\text{We get } r = -2.$$

$$\text{so, required point of intersection is } T(1, -2, 7).$$

$$\text{Hence, PT} = 7.$$

3. Ans (1)

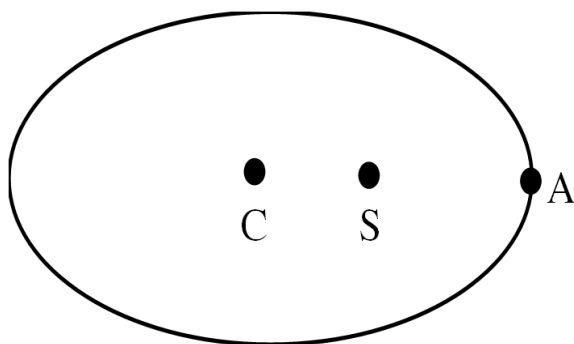
$$\operatorname{Re}(z) = \frac{3 - 6\cos^2\theta}{1 + 9\cos^2\theta} = 0$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

$$\text{Hence, } \sin^2 3\theta + \cos^2 \theta = 1.$$

4. Ans (3)

Given C(3, -4), S(4, -4)



and A(5, -4)

Hence,  $a = 2$  &  $ae = 1$

$$\Rightarrow e = \frac{1}{2}$$

$$\Rightarrow b^2 = 3.$$

$$\text{So, E : } \frac{(x-3)^2}{4} + \frac{(y+4)^2}{3} = 1$$

Intersecting with given tangent.

$$\frac{x^2 - 6x + 9}{4} + \frac{m^2 x^2}{3} = 1$$

Now,  $D = 0$  (as it is tangent)

$$\text{So, } 5m^2 = 3.$$

5. Ans (5)

$$\begin{aligned} I &= 2 \int_0^{\pi/2} \sin^3 x e^{-\sin^2 x} dx \\ &= 2 \int_0^{\pi/2} \sin x e^{-\sin^2 x} dx + \\ &\quad \int_0^{\pi/2} \underbrace{\cos x}_I \underbrace{e^{-\sin^2 x} (-\sin 2x)}_{II} dx \\ &= 2 \int_0^{\pi/2} \sin x e^{-\sin^2 x} dx + \left[ \cos x e^{-\sin^2 x} \right]_0^{\pi/2} \\ &\quad + \int_0^{\pi/2} \sin x e^{-\sin^2 x} dx \\ &= 3 \int_0^{\pi/2} \sin x e^{-\sin^2 x} dx - 1 \\ &= \frac{3}{2} \int_{-1}^0 \frac{e^\alpha d\alpha}{\sqrt{1+\alpha}} - 1 \quad (\text{Put } -\sin^2 x = t) \\ &= \frac{3}{2e} \int_0^1 \frac{e^x}{\sqrt{x}} dx - 1 \quad (\text{put } 1 + \alpha = x) \\ &= \frac{3}{2e} \int_0^1 \underbrace{e^x}_I \underbrace{\frac{1}{\sqrt{x}}}_{II} dx - 1 \\ &= 2 - \frac{3}{e} \int_0^1 e^x \sqrt{x} dx \end{aligned}$$

$$\text{Hence, } \alpha + \beta = \boxed{5}$$

6. Ans (2)

$$t^4 - t^3 - 4t^2 - t + 1 = 0, e^x = t > 0$$

$$\Rightarrow t^2 - t - 4 - \frac{1}{t} + \frac{1}{t^2} = 0$$

$$\Rightarrow \alpha^2 - \alpha - 6 = 0, \alpha = t + \frac{1}{t} \geq 2$$

$$\Rightarrow \alpha = 3, -2 \text{ (reject)}$$

$$\Rightarrow t + \frac{1}{t} = 3$$

$\Rightarrow$  The number of real roots = 2

7. **Ans ( 2 )**

$$\int e^{-y} dy = \int e^{\alpha x} dx$$

$$\Rightarrow e^{-y} = \frac{e^{\alpha x}}{\alpha} + c \quad \dots(i)$$

Put  $(x,y) = (\ln 2, \ln 2)$

$$\frac{-1}{2} = \frac{2^{\alpha}}{\alpha} + C \quad \dots(ii)$$

Put  $(x,y) \equiv (0, -\ln 2)$  in (i)

$$-2 = \frac{1}{\alpha} + C \quad \dots(iii)$$

(ii) - (iii)

$$\frac{2^{\alpha} - 1}{\alpha} = \frac{3}{2}$$

$$\Rightarrow \alpha = 2 \text{ (as } \alpha \in \mathbb{N})$$

8. **Ans ( 924 )**

$$N = 2^{10} \times 5^{10} \times 11^{11} \times 13^{13}$$

Now, power of 2 must be zero,

power of 5 can be anything,

power of 13 can be anything.

But, power of 11 should be even.

So, required number of divisors is

$$1 \times 11 \times 14 \times 6 = 924$$

9. **Ans ( 832 )**

$$B - C \equiv \{7, 13, 19, \dots, 97, \dots\}$$

$$\text{Now, } n^2 - n \leq 100 \times 100$$

$$\Rightarrow n(n-1) \leq 100 \times 100$$

$$\Rightarrow A = \{1, 2, \dots, 100\}.$$

$$\text{So, } A \cap (B - C) = \{7, 13, 19, \dots, 97\}$$

$$\text{Hence, sum} = \frac{16}{2}(7 + 97) = 832$$

10. **Ans ( 2020 )**

$$A^n = \begin{bmatrix} 1 & n & \frac{n^2+n}{2} \\ 0 & 1 & n \\ 0 & 0 & 1 \end{bmatrix}$$

So, required sum

$$= 20 \times 3 + 2 \times \left( \frac{20 \times 21}{2} \right) + \sum_{r=1}^{20} \left( \frac{r^2 + r}{2} \right)$$

$$= 60 + 420 + 105 + 35 \times 41 = 2020$$